DUAL DECOMPOSITION

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Outline

1. Dual decomposition
2. Network utility maximization

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2. Network utility maximization

Dual ascent method

Instead of solving the primal problem, we solve the dual one.
Consider the problem

$$\min_x f(x) \quad \text{subject to} \quad Ax = b$$

The Lagrangian

$$L(x, u) = f(x) + u^T(Ax - b),$$

where u_i is the multiplier associated with i-th constraint.

Dual ascent method

Define dual function

$$g(u) = \min_x L(x, u)$$

(Sub)gradient: $\partial g(u) = Ax - b$, where $x \in \operatorname{argmin}_z f(z) + u^T Az$
Dual problem is given by $\max_u g(u)$.
Dual (sub)gradient ascent:

1. $x^{(k)} = \operatorname{argmin}_x f(x) + (u^{(k-1)})^T Ax$
2. $u^{(k)} = u^{(k-1)} + t_k(Ax^{(k)} - b)$

for $k = 1, 2, 3, \ldots$. Step sizes t_k are chosen in standard ways.

Dual decomposition

Consider

$$\min_x \sum_{i=1}^{n} f_i(x_i) \quad \text{subject to} \quad Ax = b$$

where $x = (x_1, \ldots, x_n) \in \mathbf{R}^n$. $A \in \mathbf{R}^{m \times n}$.

- A large-scale problem that the centralized method cannot be applied

The minimization $\min_x f(x) + u^T Ax$ decomposes into n sub-problems:

$$x_i := \text{argmin}_{x_i} f_i(x_i) + u^T A_i x_i, \quad i = 1, \ldots n,$$

where A_i is the i-th column of A.

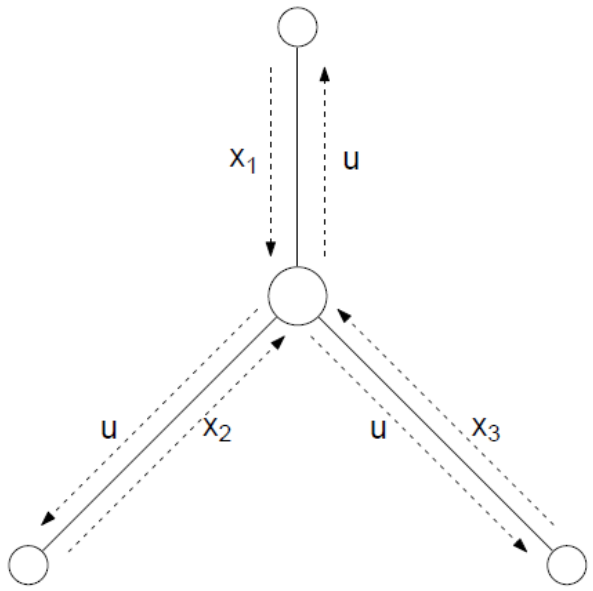
Dual decomposition

Dual decomposition algorithm: repeat for $k = 1, 2, 3, \ldots$

1. $x_i^{(k)} = \operatorname{argmin}_{x_i} f_i(x_i) + (u^{(k-1)})^T A_i x_i,\ i = 1, \ldots, n$
2. $u^{(k)} = u^{(k-1)} + t_k(\sum_{i=1}^n A_i x_i^{(k)} - b)$

Can think of these steps as:

- **Broadcast**: send u to each of the B processors, each optimizes in parallel to find x_i
- **Gather**: collect $A_i x_i$ from each processor, update the global dual variable u



Inequality constraints

Consider

$$\min_x \sum_{i=1}^{b} f_i(x_i) \quad \text{subject to} \quad \sum_{i=1}^{n} A_i x_i \leq b$$

Projected subgradient method for the dual problem:

1. $x_i^{(k)} = \operatorname{argmin}_{x_i} f_i(x_i) + (u^{(k-1)})^T A_i x_i,\ i = 1, \ldots, n$
2. $u^{(k)} = \left(u^{(k-1)} + t_k \left(\sum_{i=1}^{n} A_i x_i^{(k)} - b \right) \right)_+$

where u_+ denotes the positive part of u, *i.e.*, $(u_+)_i = \max\{0, u_i\}$.

Dual decomposition (cont.)

- We can extend the formulation to B blocks of x

$$\min_x \sum_{i=1}^{B} f_i(x_i) \quad \text{subject to} \quad Ax = b$$

  where $x = (x_1, \ldots, x_B) \in \mathbf{R}^n$ divides into B blocks of variables, with $x_i \in \mathbf{R}^{n_i}$. $A = [A_1, \ldots, A_B]$, in which $A_i \in \mathbf{R}^{m \times n_i}$.
- The minimization problem $\min_x f(x) + u^T Ax$ is decomposed into B sub-problems:

$$x_i := \operatorname{argmin}_{x_i} f_i(x_i) + u^T A_i x_i, \quad i = 1, \ldots, B$$

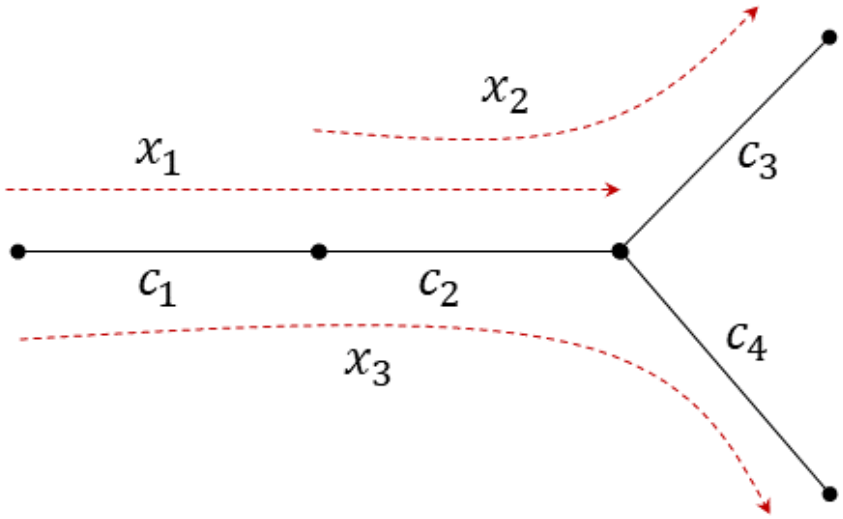
Outline

1 Dual decomposition

2 Network utility maximization

Network utility maximization

$$\begin{aligned}
\max\ & U_1(x_1) + U_2(x_2) + U_3(x_3) \\
\text{s.t. } & x_1 + x_3 \le c_1 \\
& x_1 + x_2 + x_3 \le c_2 \\
& x_2 \le c_3 \\
& x_3 \le c_4 \\
& x_1, x_2, x_3 \ge 0.
\end{aligned}$$



$R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the routing matrix.

Rewrite the constraints: $Rx \le c$ and $x \ge 0$.

Utility function

α-fair family: $U_i(x_i) = \begin{cases} w_i \frac{x_i^{1-\alpha}}{1-\alpha} \text{ if } \alpha > 0, \alpha \neq 1 \\ w_i \log(x_i) \text{ if } \alpha = 1 \end{cases}$

- $\alpha = 1$: weighted proportional fairness
- $\alpha = 2$: weighted harmonic-mean fairness
- $\alpha = \infty$: weighted max-min fairness

Lagrangian of NUM

Equivalent problem:

$$
\begin{aligned}
\min. \; & -U_1(x_1) - U_2(x_2) - U_3(x_3) \\
\text{s.t. } & x_1 + x_3 \le c_1 && : p_1 \\
& x_1 + x_2 + x_3 \le c_2 && : p_2 \\
& x_2 \le c_3 && : p_3 \\
& x_3 \le c_4 && : p_4 \\
& x_1, x_2, x_3 \ge 0.
\end{aligned}
$$

Lagrangian:

$$
\begin{aligned}
L(x,p) &= -U_1(x_1) - U_2(x_2) - U_3(x_3) + p_1(x_1 + x_3 - c_1) + p_2(x_1 + x_2 + x_3 - \\
&\quad + p_3(x_2 - c_3) + p_4(x_3 - c_4) \\
&= \big(- U_1(x_1) + (p_1 + p_2)x_1 \big) + \big(- U_2(x_2) + (p_2 + p_3)x_2 \big) \\
&\quad + \big(- U_3(x_3) + (p_2 + p_4)x_3 \big) - p_1 c_1 - p_2 c_2 - p_3 c_3 - p_4 c_4
\end{aligned}
$$

Dual function

Dual function is given by:

$$D(p) = \min_{x \geq 0} L(x, p).$$

Let's $q_1 = R_1^T p = p_1 + p_2$, $q_2 = R_2^T p = p_2 + p_3$, and $q_3 = R_3^T p = p_1 + p_2 + p_4$ which are the sum of link prices along the flow i.

$$q = R^T p.$$

$$x_i = arg \min_{x_i \geq 0} -U_i(x_i) + q_i x_i = \max(0, U'^{-1}(q_i^{(k-1)})),$$

Question: Derive x_i when $\alpha = 1, 2$?

Dual problem is given by:

$$\max_p D(p)$$
$$\text{s.t. } p_1, p_2, p_3, p_4 \geq 0.$$

At step k of the dual gradient ascent algorithm,

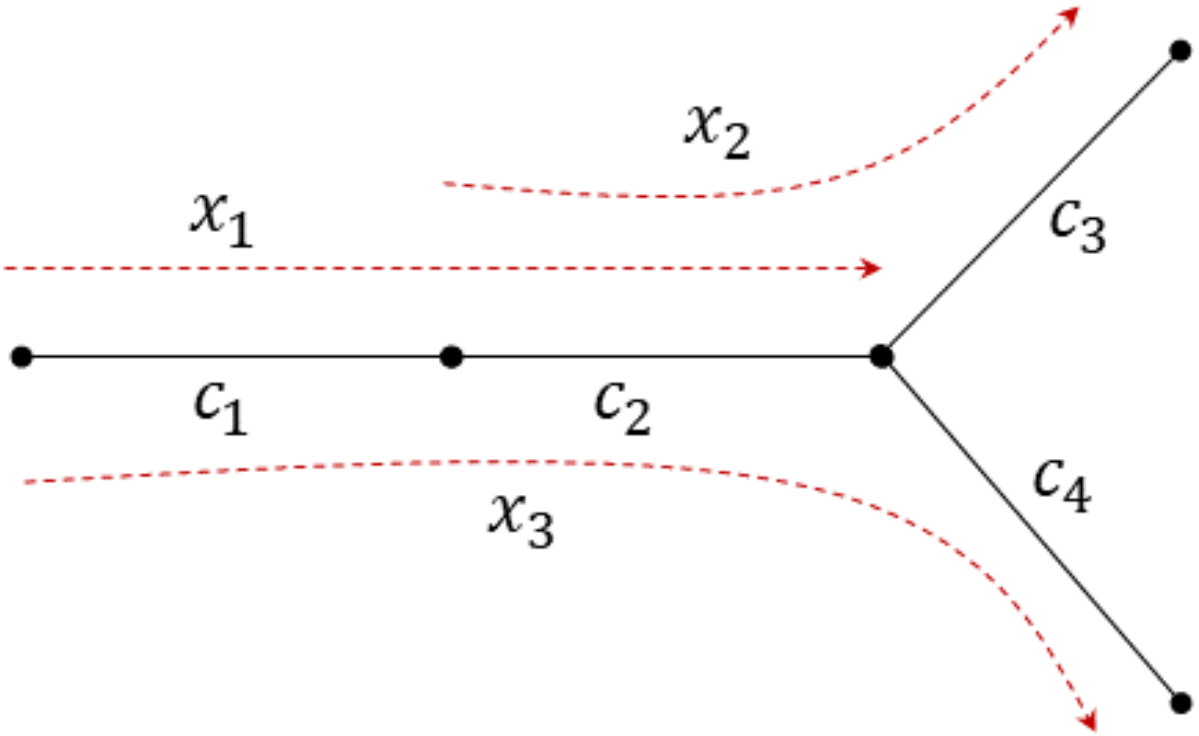
- $x_i^{(k)} = [U'^{-1}(q_i^{(k-1)})]_+$,
- $p^{(k)} = \max(0, p^{(k-1)} + t_k \nabla_p D(p))$, or
  $p_l^{(k)} = \max(0, p_l^{(k-1)} + t_k(y_l^{(k)} - c_l))$,

where y_l is the sum of rates of all the flows on link i

Interpretation

- Price update p_l follows queueing update of link l.
- q_i is the sum of prices of links along the flow i.
  - For loss-based TCP, it is measured by loss packets of the flow i.
  - For delay-based, it is the sum of delay to deliver the packet from the source to destination.

Write the code to calculate the rate allocation in following network.



Generalize the NUM model for an arbitrary network.
Extension: the NUM model for multipath-flow network.

References

- Chiang *et al.*: Layering as Optimization Decomposition: A Mathematical Theory of Network Architectures